

Automatic learning of user profiles — towards the personalisation of agent services

S J Soltysiak and I B Crabtree

Agent technology is able to provide increasingly more services for individuals, groups and organisations. Services such as information finding, filtering and presentation, service/contract negotiation, and electronic commerce are now possible. User profiling is fundamental to the personalisation of this technology. This paper describes experimental work conducted to investigate user profiling within a framework for personal agents. In particular, investigations were aimed at discovering whether user interests could be automatically classified through the use of several heuristics. The results highlighted the need for minimal user feedback, and the need to consider the implications for the role of machine learning in user profiling.

1. Introduction

Agent technology has undoubtedly produced a large impact upon computing during the last few years (see Nwana [1] for a review). Coupled with the growth of the Internet, agents are often regarded as a necessary and important feature of the near-future information-rich society. Agents which have been developed for the Internet have addressed many tasks, such as searching for information [2], and summarisation, filtering and presentation of information [3]. Further agents have been proposed which proactively recommend information to a user based on collaborative filtering [4], or indeed introduce people who share similar interests [5, 6]. Most, if not all, of these systems rely on the agent having some knowledge of the user. The inclusion of user information is rapidly becoming a key area, not only for agent technology, but also for the Internet in general, as demonstrated by the recently proposed open profiling standard [7, 8]. User profile information will undoubtedly play an ever-increasing role as digital libraries, electronic commerce, and personalised applications become more widespread.

This paper focuses upon the role of user profiling within the context of personal agents. We consider personal agents to be software capable of operating autonomously in order to provide timely and relevant information for an individual. Examples¹ are GRAPEVINE, an interest matchmaking agent, RADAR, an information presentation agent, and PANDORA, an agent that suggests topics that may be of interest to a user. These agents have been operational for some time within a test bed for the development of a framework for personal agents. Central to this framework is a user profile, which describes the interests of a user, and is available to all of the user's personal agents. We believe

there is no question that user profiles are vitally important, but furthermore that these profiles should be acquired automatically with minimal user input. That is not to say the user does not have the ability to review and correct/refine their profile, but that the majority of the effort involved in constructing the profile is performed by the user's profiler agent. After all, what use is a software agent that constantly requires the user to instruct it!

The aim of the work presented in this paper was to investigate the automatic classification of user interests to reduce the need for user feedback. We used a machine-learning algorithm to generate a set of interests, and then applied a number of heuristics to automatically classify the interests. The remainder of this paper is organised as follows. Firstly, we briefly review user profiling. We then describe the approach taken in our profiling work, and present the results of an experimental programme using the personal agent framework test bed. Finally, we discuss the wider implications of the results of our work.

2. Current user profiling

User profiling within agent technology has to date concentrated upon profiles for specific agent tasks, typically information management related, and often involving the World Wide Web (WWW). Search agents, browsing assistants and recommendation agents are examples. Such agents are often able to learn a profile of the user, while others require the user to specify their interests. A more detailed review of user profiling can be found in Soltysiak and Crabtree [9].

2.1 Learning user profiles

The WWW browsing assistant Syskill & Webert constantly observes the user and suggests which WWW pages are of interest [10]. Syskill & Webert learns the user's profile, and uses this to guide its suggestions for interesting WWW pages. Initially a user supplies Syskill & Webert with the name of the topic, and the URL of an index of WWW pages for that topic. Syskill & Webert represents interests as keyword vectors. As users browse the WWW they can manually rate pages as good or bad to give Syskill & Webert positive (interesting) and negative (uninteresting) training examples. A naive Bayes classifier is used to revise the user profile (other learning algorithms were evaluated but none performed as successfully as the Bayes classifier²). The updated user profile is then used to compare documents linked from the current page to suggest which are of interest to the user and should be followed.

Autonomy's AgentWare³ agent can accept textual descriptions of interests (keywords, phrases or a set of documents) and then applies neuro-fuzzy pattern-matching to identify the key concepts within the document in order to build up a description for the user's interests. A single 'agent' represents user interests as some form of encoded concept representation, which can be shared between agents within the Autonomy AgentWare suite. Autonomy's agents can also determine user profiles by observing the users' actions (assuming the users read items in which they are interested).

2.2 Representing user profiles

Typically, when a user marks a page as interesting (or uninteresting), or the user visits a particular page, an agent performs some analysis of that page in order to determine what feature(s) caused the user to be interested (or disinterested) in it.

Keyword extraction using TF-IDF [11] is a common technique for identifying the relevant keywords within the text. Each word has an information measure calculated on the basis of the frequency of the word in the document (term frequency, TF) and the frequency of the same word in the entire set of documents to be processed (inverse document frequency, IDF). Each document is then represented as a vector of keywords which can be stored either as an exemplar for later comparison, or as a training instance (positive or negative) for a learning algorithm to discover what does and does not interest this user. Alternatively,

² No single algorithm was found to be superior in their study, as performance was related to the domain of application. However, the naive Bayes classifier was selected because of its all-round performance, simplicity and fast execution.

³ <http://www.agentware.com/main/agent/index.html>

documents can be represented as Boolean feature vectors, where the presence or absence of a word is denoted [10].

Ultimately the user's profile will contain those features which categorise the interesting pages (and perhaps the uninteresting pages too). Such a profile is inherently data-driven — little, if any top-down user input is provided to guide or assist the construction of the profile. Although the user must specify which pages are interesting and which pages are not interesting, the user does not have direct or explicit involvement in the construction of the profile. Nonetheless, there is a high level of user supervision of the profile learning process since the user must supply an initial keyword description, or provide examples of good or bad pages. Such approaches are commonplace within agent systems, and can produce accurate profiles that further enhance that agent's performance.

3. Our approach to user profiling

We are concerned with the automatic generation of user interest profiles with minimal user input. We based our approach around the electronic documents a user reads or writes. Specifically we are interested in the e-mails they send or receive, and the WWW pages they browse. We take these documents and generate document vectors that describe them. The vectors are then used as the basis of grouping documents into clusters, the intention being that a sufficiently large cluster of documents will represent a user's interest.

Some of our earlier experimentation [12] demonstrated the suitability of this approach, but highlighted a number of issues that we investigate here. In our earlier work we used standard information retrieval (IR) techniques based around single keyword extraction and TF-IDF⁴ for documents. The results, although promising, suggested that simple keyword extraction techniques were not sufficient to produce high-quality interest clusters. We found that single keywords lose too much contextual information and the resulting clusters do not appear to be particularly good.

The experiments reported in this paper continued to use the same clustering algorithm as we believe that it provides a solid foundation for learning user profiles. In other recent work we investigated several techniques for the extraction of important and meaningful keywords and phrases from documents in order to derive higher quality document descriptions [13]. One of the techniques investigated was ProSum [3], which is a text summariser that can additionally provide a set of keywords that describe a document.

⁴ Our earlier work used a corpus of 18 million words of the English language rather than the strict TF-IDF approach of simply using the complete set of documents to be processed.

ProSum is a sentence-based text summariser, which derives a weight for each sentence according to the number of words in a sentence which appear in other sentences. ProSum incorporates other heuristics which consider the separation of sentences (e.g. whether sentences are linked through words like ‘however’), and can also exploit any HTML structure that may be present in the document. The experiments showed that ProSum generated the best document descriptors; we therefore used ProSum to generate the document vectors. These experiments are described in detail in Davies and Crabtree [13].

Figure 1 shows the overall procedure we have adopted for clustering. The primary learner generates a set of interest clusters, while the secondary learner performs the classification of the interest clusters, which is the focus of this paper. For more details of the primary learning process, see Crabtree and Soltysiak [12].

3.1 The primary learner — generating interest clusters

Users of the personal agent framework have been able to forward e-mails and WWW pages to their agent for some time in order to automatically determine their interest profiles. The documents were preprocessed using ProSum to extract keyphrases, and then document vectors were generated to serve as input to the clustering algorithm. The basic procedure we adopted is as follows:

- collect documents for clustering and generate the keyword/keyphrase information values,
- apply the clustering algorithm to these documents to produce interest clusters (we also included any documents from previous iterations, which had not yet been clustered).

3.2 The secondary learner — classifying interest clusters

Having generated a set of clusters, the next stage was to classify them to determine which were interests and which were not. Fitting clusters to an existing profile is an

important part of our approach — but to do this we must be able to determine which of the latest clusters are sufficiently close to existing interests such that they should be added automatically. Classification of interests is a two-stage process, the first using a basic similarity measure, and the second based on more detailed characteristics of existing interests.

Initially we generated a matrix of the similarity of all new clusters with all existing interests and used this as the basis of fitting clusters to interests. We used two similarity-based heuristics for initially deciding whether a new cluster should be considered an interest or not:

- if the new cluster is sufficiently similar to an existing cluster it is automatically added as good (or marked bad if the existing cluster was bad) — we set a threshold to determine whether this cluster is sufficiently close,
- if the new cluster is not sufficiently close to existing clusters, the new cluster is compared with all known good clusters and bad clusters to determine which it is closer to (i.e. to decide whether the likely classification of the cluster is good or bad) — for each of the existing good interests g and bad clusters b , the difference of the weighted sum of the similarities is calculated as:

$$\sum_g \left(\frac{\text{ndocs}_g}{\text{Ndocs}_g} \times \text{sim}(c, g) \right) - \sum_b \left(\frac{\text{ndocs}_b}{\text{Ndocs}_b} \times \text{sim}(c, b) \right)$$

where c is the current cluster under consideration, Ndocs_g is the number of documents comprising the existing good interest g , and $\text{sim}(c, g)$ is the similarity between the existing interest g and the current cluster c . Likewise the subscript b represents existing bad clusters. Weighting is scaled depending upon how large each good/bad cluster is (Ndocs). The sign of this difference is used to determine whether the cluster is likely to be an interest (positive value) or not (negative value).

The thresholds we employed allow the profiler to automatically add the cluster as good or bad if the

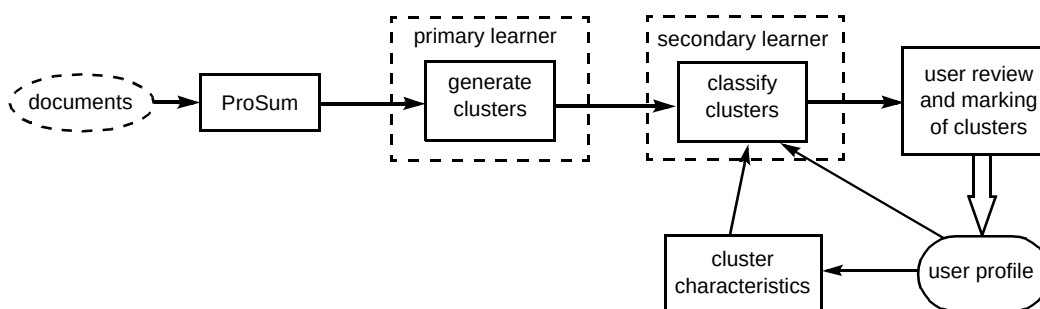


Fig 1 Overview of the clustering process.

difference exceeds the relevant threshold, or to remain unsure if the difference value lies between the two thresholds. In this latter case the cluster is presented to the user who is then asked to specify whether it is an interest or not.

Figure 2 shows an example of the screen presented to a user for them to mark their interests. The figure shows one existing interest (adaptive personal agents) and one bad cluster.

Although there are actually many more interests in the user's profile, these are displayed because they have new clusters that are similar to them, but not sufficiently similar to warrant automatic classification of them. At the bottom is a new cluster that is not close to any existing interest and thus has to be marked by the user.

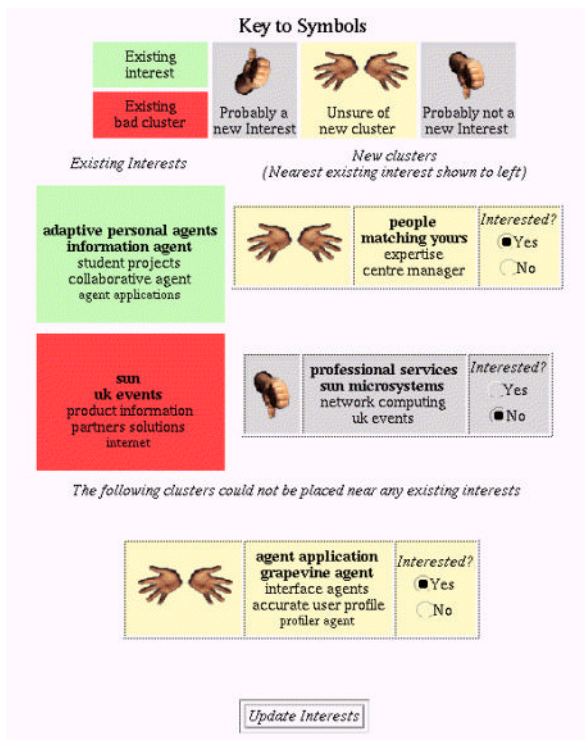


Fig 2 User display for marking of clusters.

The purpose of these heuristics is primarily to ensure that the profiler can perform as much of the classification of interest clusters as possible, thereby requiring the user to provide minimal input to the profile generation process. Of course, when the profiler first generates a set of clusters for a user, it has no reference point to which to try to fit clusters, and so it will automatically request the user to state which are interests and which are not. Nevertheless, as time progresses, the profiler becomes more proficient at automatically classifying the new clusters, so that the user has very little to do.

4. Experimental programme

Our experimental programme concentrated upon the evaluation of different metrics for deciding whether clusters were good or bad. These augment the heuristics described above in section 3.2 which auto-fitted clusters to previous clusters which had been marked interesting or not. We further varied a number of other parameters to assess their impact upon the classification process.

4.1 Cluster characteristics

Once documents were clustered by the primary learner, users were asked to mark them as either interesting or uninteresting. This data provided a baseline from which we calculated several features for each cluster.

We analysed all the good and bad clusters generated for a user over the entire period clustering occurred, and calculated several metrics for each cluster:

- cluster size (the number of documents forming the cluster) — interests are expected to produce many documents, therefore a large cluster size may be indicative of interests,
- cluster age (the number of days between the earliest and latest documents in a cluster) — interests are expected to persist over time, therefore if a cluster has documents spanning a large number of days it may be indicative of an interest,
- mean time per document (for WWW pages only) — one might expect good pages to be read for longer than bad pages,
- mean document size — this may be correlated with the mean time per document, and may suggest interest in a document or cluster,
- mean number of hyperlinks in clusters — e-mails with references to WWW pages, which are then followed, would be expected to show possible areas of interest,
- mean number of links followed — a cluster with a large number of followed links may suggest the cluster is comprised of search results, or pages which act as a chain through which the user navigates to arrive at a good page.

By calculating these metrics we hoped that there may be some characteristics of good and bad clusters that would help to automatically classify new clusters as they were learned. For example, if good clusters have a longer 'age' and larger cluster size than bad clusters, any new cluster with these characteristics could be regarded as good. These characteristics can then be applied to complement the initial classification heuristics of section 3.2.

4.2 Other parameters

We investigated other aspects of the profile-learning process, covering the method used to select documents for the clustering algorithm, and parameters for the algorithm itself.

Document selection

The clustering process outlined in section 3.1 was also varied to determine the most appropriate method. We evaluated several alternative procedures for selecting the documents to be used as the basis for clustering:

- the sets of documents generated in one week were clustered and compared with clusters generated in previous weeks to track interests,
- the sets of documents generated up to the start of the experiments were clustered as the basis of the initial profile — we then adopted the same weekly approach as above for subsequent clustering,
- all the historical documents generated by a user were clustered each week, and the clusters compared with those generated during the previous week,
- all documents generated during a six-week time period were clustered — this procedure was continued for each week, thus acting as a six-week-wide sliding time window.

The different volumes of documents available are likely to affect the performance of the clustering algorithm. Previously we observed that the algorithm required large volumes of documents to perform accurately [12], and we therefore expected the third and fourth methods to produce better results.

Minimum cluster size

We varied the minimum cluster size to assess the impact upon the clusters produced. By requiring more documents to form a cluster it was hoped that the resultant clusters would be more indicative of the user's interests. One would expect a user to generate more documents for topics that were relevant and interesting to them than for topics that were not. Moreover, it is likely that fewer clusters would be produced, thus requiring less user marking should the user be presented with clusters that could not be automatically classified.

Keyword/phrase information ratings

A final investigation compared the calculation of the keyword information measure using TF and TF-IDF. Although ProSum generates a set of keywords/phrases that are indicative of the document, these do not have an

associated information measure. We compared the results of clustering using the straightforward term frequency of the keyword/phrase in the document with that from the TF-IDF technique. The purpose was to determine whether, by using ProSum, we could avoid the need for calculating TF-IDF values.

5. Experimental results

5.1 Cluster characteristics

The results of the heuristics were plotted as scatter diagrams to see whether there were any distinct groupings of good and bad clusters. Unfortunately, this was not the case. For example, Fig 3 shows three such characteristics — the age of the cluster (in days), the size of the cluster (in documents) and the mean time spent on the document (in seconds). The plot shows that both good and bad clusters are positioned throughout the space with no distinct boundaries between good and bad clusters. Plots of most characteristics except cluster size failed to show distinct groupings.

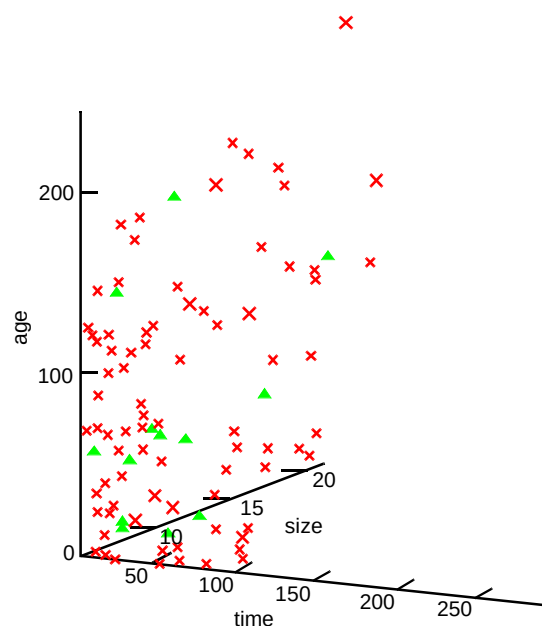


Fig 3 Sample cluster characteristics for good (triangle) and bad (cross) clusters. Axes show age of clusters, size of clusters and average time spent on documents in the clusters.

5.2 Cluster size

Variation of the minimum number of documents required to form clusters helped to produce improvements in the classification characteristics of clusters. Figure 4 shows that varying the cluster size does appear to have an effect in improving these characteristics. By increasing the minimum number of documents required to form a cluster, we see an improvement in the discrimination between good

and bad clusters. However, cluster size is not necessarily an ideal characteristic, since search engine results or 'server error' messages could appear many times in the document set to be clustered. Nonetheless, the results with a larger minimum cluster size do show improved discrimination between good and bad clusters. Also, with a larger minimum cluster size, fewer clusters were produced, which reduces the burden on the user to mark these clusters.

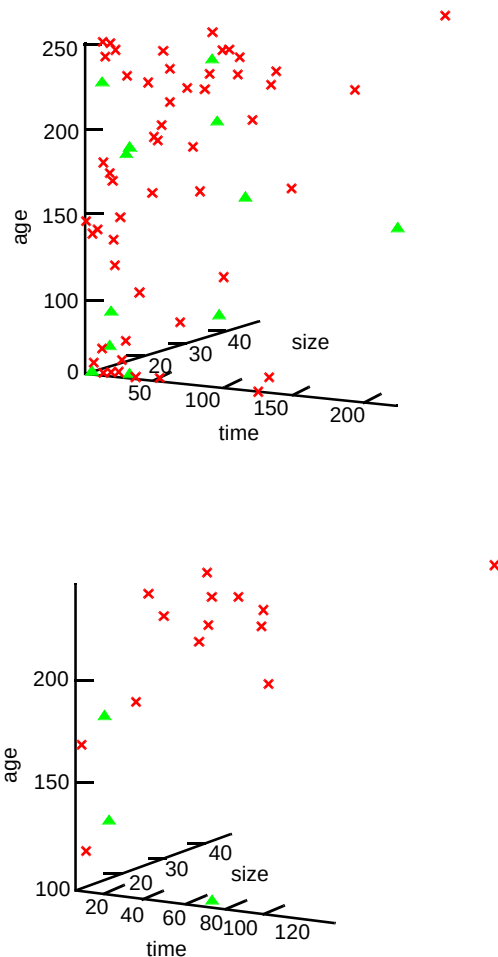


Fig 4 Clusters produced for minimum cluster size of four documents (top) and eight documents (bottom). Good clusters are triangles, bad clusters are crosses.

5.3 Information measure

We also compared the results of clustering using TF-IDF values and straight TF values for the key phrases. TF-IDF produces significantly fewer clusters, and their characteristics appear to show a clearer separation of good and bad clusters than the TF results. This is undoubtedly a consequence of the fact that TF-IDF inherently considers each document in relation to all others to be clustered; we therefore get a clearer indication of the relative importance of each keyword/phrase within a document vector. Thus the advantages of reduced processing time by simply calculating term frequencies for each keyword are

outweighed by the improved results through using TF-IDF ratings.

5.4 Clustering process

The different procedures used for clustering the documents also produced some interesting results. It would appear that the most suitable method is to take the entire set of documents for clustering, which confirms earlier results [12]. However, the processing overheads of this may be prohibitive if clustering is performed on a weekly or perhaps more frequent basis. A compromise is to use the six-week-wide sliding time window approach.

The results of the sliding window approach show that many bad clusters tend to group together. This is to be expected, since many of the same documents appear in more than one week. Figure 5 shows a plot of several statistics from the time-series experiment. It is clear that as the learning process continued over time, the secondary learner automatically classified most of the clusters, leaving the user with minimal marking, as can be seen in Fig 5, where typically 70% of the new clusters are automatically classified. The increase in the number of documents available provided the secondary learner with many examples of good and bad clusters, from which it was able to automatically classify many of the clusters.

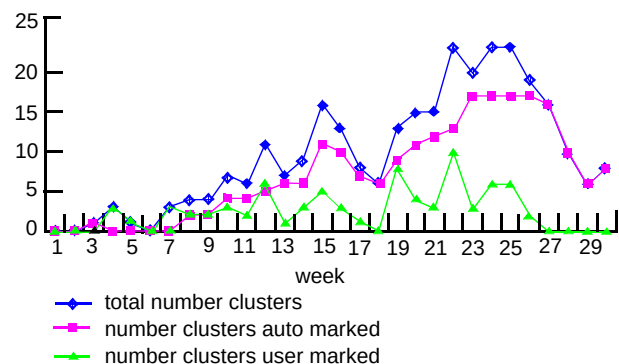


Fig 5 Time-series experiment clustering statistics.

Since many clusters appear in consecutive weeks (because each document is available for clustering for six weeks) they are close to the same cluster in the previous week. Thus when the initial classification heuristics of section 3.2 are applied, the user only has to mark any new clusters which appear, and for the last six weeks of the experiment the user did not have to mark any clusters. With this approach, the need for secondary characteristics appears minimal — we can simply classify interests as good or bad according to the similarity to previous clusters. Of course, we still require an initial classification for a cluster, which must be provided by the user.

6. Discussion

The experimental results demonstrated that it was difficult to automatically classify all clusters using a purely data-driven approach. However, we found that using a sliding time-window approach and having larger minimum cluster sizes did significantly reduce the effort required of the user.

One possible explanation of being unable to identify suitable discriminatory features may be the representation used. TF-IDF document vectors may be at too low a level to be able to draw meaningful hypotheses about good and bad clusters. Essentially we have been learning at the 'syntactic' level, and we may need to consider learning at a higher semantic or conceptual level in order to classify clusters more accurately. Latent semantic indexing [14] is one technique that may be able to provide the richer representation we require. Alternatively we could consider the use of a thesaurus or other such source of data such as WordNet [15].

The characteristics (such as number of documents, age of the cluster, and so on) by which we have tried to examine the clusters are those that are meaningful to us, humans, rather than those that are meaningful to the learner. Since the learner 'understands' document vectors, can we realistically expect to distinguish between clusters in some other n-dimensional space? If we draw an analogy with neural networks, they are able to learn how to classify inputs into outputs, yet their knowledge is coded within the weighted connections; it is simply not accessible directly to the outside world. Perhaps we have therefore approached the problem of classification too simplistically. On the other hand, perhaps it is unrealistic to expect any machine-learning algorithm to be able to determine whether a cluster is an interest or not — user intervention may well be the only method to obtain a true classification. With a machine-learning approach we can either provide good and bad examples for the primary learner to learn from, or we can provide feedback for the classifier to refine its ability to automatically add clusters.

Although we have not been completely successful in automatically determining a user's interest profile, our approach has produced significant results. The most appropriate method for learning user interests is to apply a sliding time window technique, and to use cluster similarity as the guide for cluster classification. Clearly the user must provide some initial indication to the learner as to what constitutes an interest. In the context of our goal — learning user interests with minimal user interaction — we have succeeded. Although we have not found an effective set of rules for automatic classification from the beginning, the approach we have taken allows us to present a minimal set of yes/no check-boxes to the user to update their interest profile with minimal effort.

7. Conclusion

This paper has presented experimental results for automatically classifying learned user profiles. The overriding result is that we can minimise the effort for the user in specifying interests to that of saying yes/no, but not eliminate it altogether. Many current user profiling mechanisms require substantial user guidance for the learning process by giving feedback on a document-by-document basis. As we argued in the introductory sections, any user involvement should ideally be reduced to a minimum. The goal for machine learning is therefore to reduce the need for user interaction rather than replace it.

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models of language processing.

Stuart Soltysiak obtained a BSc in Command and Control, Communications and Information Systems from Cranfield Institute of Technology (1992), followed by an MSc in Cognitive Science (1993) and PhD in Computer Science from Essex University (1997). Since joining BT in 1996, he has been working on personal software agents, including machine learning of user interests and security and privacy issues related to personal agents. Alongside his current work with software agents, and artificial intelligence in general, he retains research interests in cognition, particularly cognitive



Barry Crabtree joined BT in 1980 after completing a Physics degree at Bath University. After a number of years in software development he moved into the field of artificial intelligence (AI) and helped begin BT's systems in automated diagnosis and workforce management. He moved on to work in distributed AI, optimisation problems and adaptive systems. He is currently a technical advisor in Applied Research and Technology (ART), and has particular interest in intelligent agents and wearable computers.